

Dhruva Aher

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PROJECTS

JusticeQueue – AI Legal Triage Platform

[\(link\)](#) | justicequeuelive.vercel.app

Next.js 14, GCP Vertex AI, MongoDB Atlas, Gemini API, Upstash Redis, Firebase

- Designed a 13-step model-directed pipeline where Gemini Flash makes 8 decisions per run — selecting from 4 tool combinations, allocating Atlas Vector Search resources, evaluating evidence sufficiency, and self-critiquing recommendations — each persisted as a typed evidence document in MongoDB Atlas.
- Integrated 5 Google Cloud services (Vertex AI Gemini Pro/Flash, text-embedding-004, Cloud Logging, Cloud Monitoring) with Firebase Auth and OAuth 2.0; deployed on Vercel with Upstash Redis rate limiting and 7 custom Cloud Monitoring metrics per docket run.

Aura – Distributed Task Orchestration Platform

[\(link\)](#) | aurasys.vercel.app

Node.js, PostgreSQL, Redis, Docker

- Engineered a distributed job scheduling system in Node.js handling 20,000+ concurrent tasks with priority queues, worker crash detection, and automatic job recovery; system sustained 300 jobs/min under peak load with P95 latency of 6.5 seconds.
- Implemented safe job leasing with Redis to prevent duplicate execution across workers; added retry logic with exponential backoff, heartbeat-based crash detection, and a real-time monitoring dashboard for queue observability.
- Wrote **250 unit tests** covering concurrency failures and edge cases; validated crash recovery restoring 6,000+ jobs in under 2 seconds after complete data loss.

FlowBoard – Full-Stack SaaS Platform

[\(link\)](#)

React, TypeScript, FastAPI, PostgreSQL, Redis, WebSockets

- Built a project management web application with real-time collaborative editing, role-based access control, and multi-tenant workspace isolation using React, TypeScript, and WebSockets.
- Designed RESTful backend with FastAPI and async PostgreSQL; implemented JWT authentication, Alembic schema migrations, and optimistic UI updates via TanStack Query and Zustand.

PRBeliefs – Automated Code Review Tool

[\(link\)](#)

Python, Redis, SQLite, Docker, GitHub Actions

- Built an automated code review tool that dispatches parallel review workers via a Redis-backed async queue; outputs structured, ranked feedback directly on GitHub PRs with per-installation rate limiting.
- Designed a webhook ingestion pipeline decoupled from review processing using async queues; deployed with Docker Compose and GitHub Actions CI with automated testing on every push.

EXPERIENCE

ACM at ASU – Coordinator, Administration

Tempe, AZ | March 2026 – Present

- Coordinated GlobeHacks 2026 (ACM x GCN) end-to-end; managed Devpost configuration, real-time Discord announcements, and participant support for 250+ attendees across technical and non-technical tracks including a Google-hosted workshop session.
- Manage ACM's Discord community of 250+ members; facilitate recurring technical workshops and speaker sessions on AI, distributed systems, and industry topics attended by 30–40 students per session.

EDUCATION

Arizona State University, Grand Challenges Scholars Program

Tempe, AZ | Aug 2025 – May 2029

Bachelor of Science in Computer Science

GPA: 4.0/4.0

Coursework: Data Structures & Algorithms, Object-Oriented Programming & Data Structures, Digital Design Fundamentals, Discrete Mathematics, Calculus for Engineers II

Honors: Dean's List (Fall 2025, Spring 2026)

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, Bash

Frontend: React, Next.js, TypeScript, HTML5, CSS, Zustand, TanStack Query, Vite

Backend & APIs: Node.js, FastAPI, Flask, REST APIs, JWT Authentication, RBAC, SQLAlchemy, Alembic, OpenAI API, Google Gemini API, Vertex AI, LLM Tool Calling, MongoDB Atlas Vector Search

Databases & Systems: PostgreSQL, Redis, MongoDB, WebSockets, Distributed Systems, Linux, Git

Cloud & DevOps: AWS (EC2, ECS, S3, RDS, CloudFront), GCP (Vertex AI, Cloud Logging, Cloud Monitoring), Microsoft Azure, Docker, Terraform, GitHub Actions, CI/CD

Certifications: [AWS Cloud Practitioner](#) | [Microsoft Azure AI Essentials Professional](#)